



DEALER BEST PRACTICE SERIES

Transmission Control
Valve Test Bench

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Transmission control valve testing is typically conducted on the transmission test bench or while the valve is installed on the machine. The valve is tested for appropriate clutch pressures and clutch modulation characteristics.

Efficient and accurate valve testing is an important part of the component repair/rebuild process. A valve that performs inaccurately can result in poor truck cycle times, reduced transmission clutch life, and high fuel consumption. Other drive train components may see higher stresses and reduced life due to inaccurate transmission shift points and poor clutch pressure modulation. Additional component failures, increased repair cost, and reduced machine availability.

A control valve test bench improves testing efficiency and quality. Quick, in-shop testing capabilities provide rebuild quality and diagnostic feedback that reduces customer machine downtime, dealer repair and/or warranty labor.

Control valve test benches provide dealer and customer value by allowing the dealer to test and adjust valves rather than just replace parts when troubleshooting the valve/transmission. It also reduces transmission-related warranty costs with enhanced valve performance information.

This Best Practice describes the test equipment, test area organization and process for efficient and effective control valve testing.

2.0 Best Practice Description

The common rebuild practice for ICM and other control valves includes:

- Disassembly and cleaning
- Standard replacement and worn parts replacement
- Spring/electrical parts tests for reuse
- Assemble with original shim packs

The next steps involve installing the control valve on the rebuilt transmission, and testing/adjusting the individual pressure regulating/modulating valves to factory specifications. This process is very tedious and time consuming, providing opportunities for dropping shims and other small parts into the transmission main case and planetary/clutch groups.

The Best Practice for this procedure would include using a control valve test bench to quickly test/adjust valves, independent of the planetary group, prior to installing the valve on the transmission. The bench provides a consistent clutch-fill exercise for the control valve. It measures leakage, determines clutch pressures, and can “trace” the clutch modulation/sequence - functionally testing individual valves in the control valve package. It reduces set-up/test variables before installing the valve on the transmission for minor adjustments (if any). If a shim or other small part is dropped during adjustment, it falls on the test bench’s drain pan for easy retrieval.

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Vehicular transmission shift controls may be added to the test bench for efficient test processing. Control valves may be removed from machines in the field and tested to establish valve performance or to troubleshoot poor transmission performance. It allows for quick control testing, separate from the clutch groups. Test specifications for valve inputs (oil pressure/flow and valve output results) are retrieved from Caterpillar references and also established through dealer testing experience and dealer-to-dealer networking/sharing.

Transmissions that would typically benefit from this enhanced test bench process include:

- ICM type transmission
- Track Type Tractor (N through T Series)
- Wheel Loader (E through H Series)
- Motor Grader (G through M Series)

Please contact GLENCO, LLC, to establish complete list of applications.

3.0 Implementation Steps

- **Determine area in shop to be designated for testing.**
 - Assure testing area has supporting utilities (solvent washer and compressed air for cleaning/clearing, lighting, and electricity as needed).
 - Assure area has adequate space and access for equipment/tools, valve test adapters, and parts staging/protection/transfer.
 - Assure area has appropriate environmental contaminations control, similar to the hydraulic test bench cell or “clean room” rebuild areas.
 - Oil flow and pressure may be supplied by 1.) The hydraulic test bench, or 2.) An independent oil pump/reservoir (optional add-on for the control valve test bench).
- **Determine how current rebuild process would be effected by this capability**
 - Adjust product flow through rebuild center through:
 - Specialized control valve rebuild processing
 - Increased testing of valves coming from field
 - “Exchange” valve strategy for instant uptime
 - Determine how valves may be protected for transfer within shops to/from field.
 - Consider the most effective way test specifications would be established, organized, stored, retrieved.
- **Establish new/adjusted repair/rebuild time requirement targets as needed.**
 - Eliminate the current processes/tooling/equipment strategy.
 - Reduce test cell labor hours and turnaround time.
 - Establish new valve rebuild/test/adjust standard job segments/operations.
- **Install the layout/equipment changes into the shop**
 - This may occur in phases, as needed, such as shop arrangement for only valve testing, then possibly valve rebuild specialization.
 - Establish valve staging/protection/transport areas for effective product handling/flow.
 - Promote change feedback as installation may show new problems.

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- Train employees to use the new procedures/equipment.

- Suggest limiting the number of valve-test bench operators/technicians. Transmission rebuilders typically may not test their transmission valves. Possibly combine valve and transmission test functions, valve/pump rebuild and test functions, etc.
- Develop input/output test spec development process through technical support group.

- Establish equipment maintenance schedule/responsibility.

- Test bench operator is typically responsible for PM's and calibrations as needed.
- PM program is documented/directed through tool management program such as Tool Watch. The same program used to manage other tool calibration processes.
- Proactive PM program would include:
 - Future maintenance schedule based on time, bench hour-meter, or valve count.
 - Maintenance activity performed and signed off per control document (QC checklist).
 - Maintenance/repair history effectively stored/organized for retrieval/review.

- Review process/equipment performance once established.

- Establish and track related rebuild/test and field troubleshooting labor variance, control valve and "unexplained" transmission clutch failure rates, and transmission life. Acquire base levels and compare to performance once the valve test bench is established.
- Process change may occur in phases. Initiate valve testing/set-up first, then possibly valve rebuild specialization, then Pressure-Time (PT) pump rebuild specialization, etc.
- Establish valve staging/protected areas, and transport methods, for effective product handling/flow.

- Establish and install any adjustments as needed.

- Promote staff feedback since process changes may develop unique problems for shop.
- Monitor field diagnostic's to assure appropriate test bench utilization – non-traditional.

4.0 Benefits

- **Saves Labor** – Valve set-up and test process reduces rebuild/test labor. Valves are processed in a controlled area where loose parts are easier to install and hard to lose. Test cycles are faster without transmission spin-ups.
- **Saves Time** – Valve rebuild may be rebuilt/tested simultaneously with the transmission clutch and planetary groups. Final transmission test/troubleshooting occurs faster in the test cell, increasing cell capacity.
- **Increases job flexibility** – Testing is performed independent of the planetary/clutch group, simplifying transmission diagnostic for the rebuild and repair shops, and field service.
- **Increases quality** – Less activity on test bench sump pan reduces test oil contamination. Valve tests also provide greater quality control to major component. Field diagnostics and root cause identification are more accurate when supported by an independent valve test, resulting in the right repair the first time.

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- **Increases profits** – Valve set-up labor reductions, test cell utilization increases, and enhanced/simplified diagnostics reduce cost. Increased quality provides greater machine availability, resulting greater product/service price realization from customer.

5.0 Resources Required

- Facility

- Re-organize work area to accommodate valve test bench, incoming/outgoing valve staging/protection, solvent washer, workbench (as needed). Provide clean room environment, electrical and hydraulic utilities as needed.

- Support Equipment

- Test bench and hydraulic pump/reservoir as needed.
- Cabinet for tools/misc parts/supplies.
- Workbench/vise and general/specific task lighting as needed.
- Solvent washer and compressed air filter, regulator, and blowgun.
- Staging racks/enclosures.

- People

- Review current valve processing concerns with rebuild team.
- Establish process, layout, and equipment improvements through team.

- Training

- Define/establish/enforce new processes, roles and responsibilities with production teams.
- Train valve rebuild and testing technicians as needed.

6.0 Supporting Attachments / References

- Glenco_Test_Bench.pdf

7.0 Related Best Practices

- 1006-4.2-1029 – Power Train Valve Rebuild Strategy

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8.0 Acknowledgements

Special thanks to:
Glen Rogers
Cashman Equipment

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